

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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granularity, providing a near real-time view of labor market adjustments. By linking this data to established measures of occupational AI exposure and other variables, we can quantify the realized employment changes since the widespread adoption of generative AI.

This paper systematically presents six key facts that emerge from the data, offering an assessment of how the AI revolution is reshaping the American workforce.

Our first key finding is that we uncover substantial declines in employment for early-career workers (ages 22-25) in occupations most exposed to AI, such as software developers and customer service representatives. In contrast, employment trends for more experienced workers in the same occupations, and workers of all ages in less-exposed occupations such as nursing aides, have remained stable or continued to grow.

Our second key fact is that overall employment continues to grow robustly, but employment growth for young workers in particular has been stagnant since late 2022. In jobs less exposed to AI young workers have experienced comparable employment growth to older workers. In contrast, workers aged 22 to 25 have experienced a 6% decline in employment from late 2022 to July 2025 in the most AI-exposed occupations, compared to a 6-9% increase for older workers. These results suggest that declining employment AI-exposed jobs is driving tepid overall employment growth for 22- to 25- year-olds as employment for older workers continues to grow.

Our third key fact is that not all uses of AI are associated with declines in employment. In particular, entry-level employment has declined in applications of AI that *automate* work, but not those that most *augment* it. We distinguish between automation and augmentation empirically using estimates of the extent to which observed queries to Claude, the LLM, substitute or complement for the tasks in that occupation. While we find employment declines for young workers in occupations where AI primarily automates work, we find employment growth in occupations in which AI use is most augmentative. These findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.

Fourth, we find that employment declines for young, AI-exposed workers remain after conditioning on firm-time effects. One class of explanations for our patterns is that they may be driven by industry- or firm-level shocks such as interest rate changes that correlate with sorting patterns by age and measured AI exposure. We test for a class of such confounders by controlling for firm-time effects in an event study regression, absorbing aggregate firm shocks that impact all workers at a

4.2 Fact 2: Though overall employment continues to grow, employment growth for young workers in particular has been stagnant

Figure 4 shows overall employment trends across age groups, pooling together all occupations. Overall employment remains robust, coinciding with a low national unemployment rate in the post-pandemic period. However, Figure 4 suggests some leveling off in employment growth for young workers relative to other age groups, consistent with recent discussion of a potentially worsening job market for entry-level workers (Chen, 2025; Federal Reserve Bank of New York, 2025).

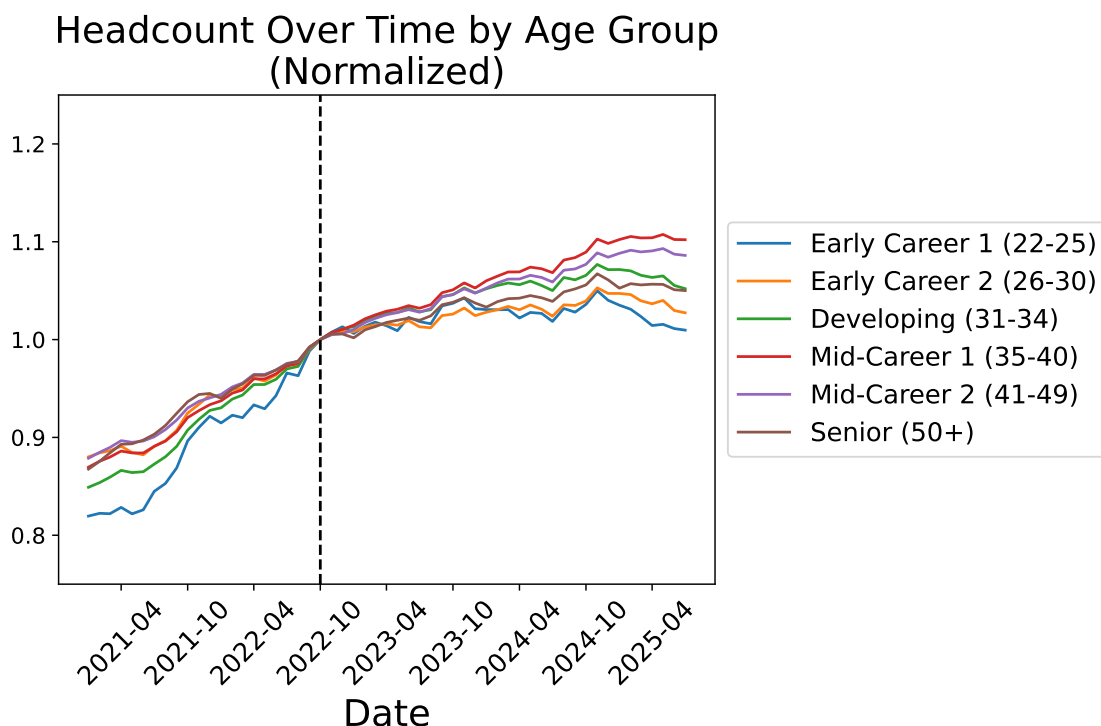


Figure 4: Employment changes by age. Including all occupations.

Figure 5 offers insight into how these trends relate to AI exposure. For each age group, employment growth from late 2022 to July 2025 was 6-13% for the lowest three AI exposure quintiles, with no clear ordering in employment growth by age. In contrast, for the highest two exposure quintiles employment for 22-25 year olds declined by 6% between late 2022 and July 2025, while employment for workers aged 35-49 grew by over 9%. These results show that declining employment in AI-exposed jobs is driving tepid overall employment growth for workers between the ages of 22 and 25.

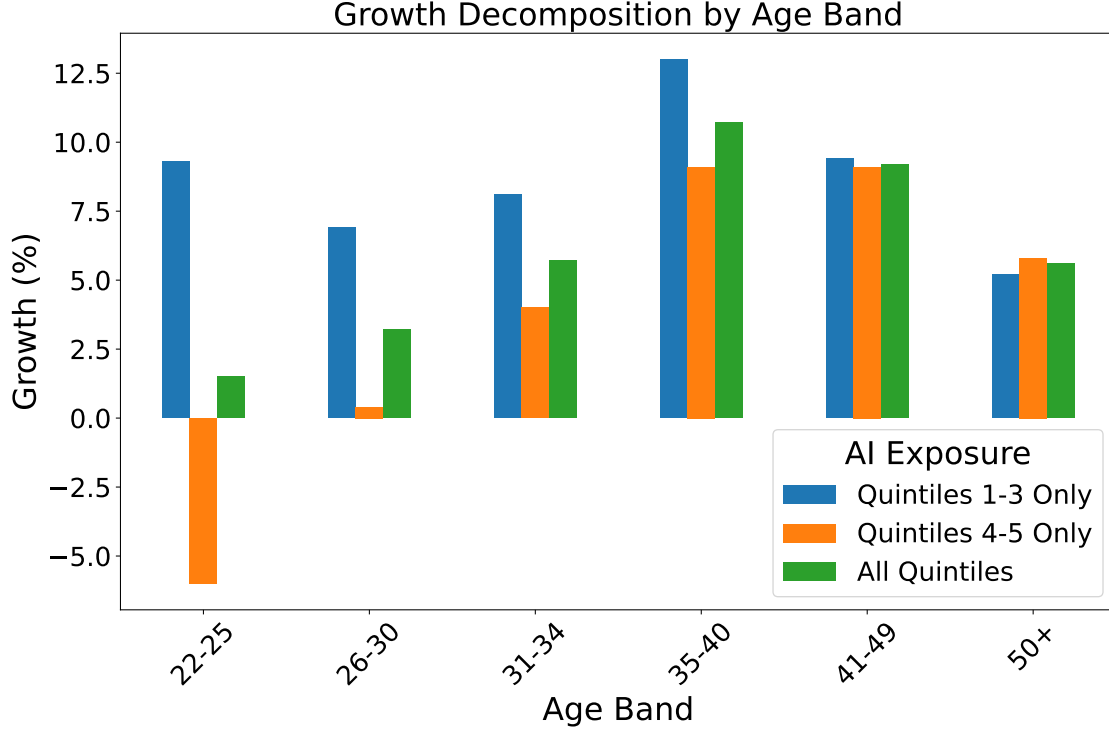


Figure 5: Growth in employment between October 2022 and July 2025 by age and GPT-4 β -based AI exposure group.

While these findings suggest divergent employment outcomes by AI exposure for young workers, we caution the trends observed in these first two facts could be driven by other changes in the US economy. Our subsequent facts evaluate the robustness of the results to alternative analyses.

4.3 Fact 3: Entry-level employment has declined in applications of AI that *automate* work, with muted changes for *augmentation*

AI exposure can either complement or substitute for labor. These may have very different implications for the labor market (Brynjolfsson, 2022).

To assess how employment patterns differ based on the complementarity or substitutability of AI with labor, we use data on generative AI usage from the Anthropic Economic Index (Handa et al., 2025). The Index provides an estimate of the share of queries that pertain to each occupation. In addition, for each task it reports estimates of the share of queries pertaining to that task that are “automative,” “augmentative,” or none of the above. We use this information as an estimate of